

Answer **all** questions in the answer booklet provided. Please start each question on a new page. Full marks are not necessarily awarded for a correct answer with no working. Answers must be supported by working and/or explanations. Solutions found from a graphic display calculator should be supported by suitable working. For example, if graphs are used to find a solution, you should sketch these as part of your answer. Where an answer is incorrect, some marks may be given for a correct method, provided this is shown by written working. You are therefore advised to show all working.

1. [Maximum mark: 26]

The following question explores features of composed trigonometric functions, such as $\sin(\sin x)$, $\sin(\sin(\sin x))$.

Suppose $S_n(x)$ denotes the function $\sin x$ composed within itself $n - 1$ times, defined for $n \geq 1$, $n \in \mathbb{Z}^+$, where $0 \leq x \leq 2\pi$.

For example, $S_1(x) = \sin x$ and $S_2(x) = \sin(\sin x)$ where $0 \leq x \leq 2\pi$.

- (a) On the same axes, sketch and label the graphs of $y = S_1(x)$ and $y = S_2(x)$. On your sketch, show the values of the intercepts with the axes. [4]
- (b) Determine the maximum value of
 - (i) $S_1(x)$; [1]
 - (ii) $S_2(x)$; [1]
 - (iii) $S_3(x)$. [1]
- (c) Find the least value of n for which the maximum value of $S_n(x)$ is less than 0.6. [3]

Consider the graph of $y = S_2(x)$.

- (d) By considering the equation $\frac{dy}{dx} = 0$, show that there are exactly two points of zero gradient, one at $x = \frac{\pi}{2}$ and one at $x = \frac{3\pi}{2}$. [6]

(This question continues on the following page)

(Question 1 continued)

The derivative $S'_n(x) = \frac{d}{dx}(S_n(x))$ can be expressed as a product of cosine functions, as follows:

$$S'_n(x) = \cos(S_{n-1}(x)) \cos(S_{n-2}(x)) \dots \cos(S_1(x)) \cos x.$$

(e) Hence, show that $S'_3(x) = \cos(\sin(\sin x)) \cos(\sin x) \cos x$. [1]

(f) Use mathematical induction to prove that for all $n \in \mathbb{Z}^+$

$$S'_n(x) = \cos(S_{n-1}(x)) \cos(S_{n-2}(x)) \dots \cos(S_1(x)) \cos x. \quad [6]$$

(g) Use l'Hôpital's rule to show that $\lim_{x \rightarrow 0} \frac{S_n(x)}{x} = 1$, for $n \in \mathbb{Z}^+$. [3]